MGT 634: NYC 311 Service Analytics

Lab Day 2

Scenario Context:

You are a data analyst at McKinsey & Company, working on a high-priority engagement for the NYC Mayor's Office. The city has provided you with 311 service request data to help them understand patterns in citizen complaints and improve city services. Your analysis will directly inform resource allocation decisions and policy recommendations for the Mayor's strategic planning team.

The Mayor is particularly concerned about:

- Response time disparities across different neighborhoods
- Seasonal patterns in service demands
- Whether certain types of complaints are indicators of broader systemic issues

Your managing partner has asked you to complete this initial data assessment to prepare for the client presentation next week.

Data:

The data are spread over a few thousand json files.

Part 1: Data Quality Assessment

The Mayor's Chief of Staff has expressed concerns about the reliability of the 311 data. Examine the dataset for missing values and data quality issues.

a) Identify which columns have missing values and calculate the percentage of missing data for each. This is not trivial as some columns may specify "missingness" in different ways. In addition, some entries may be blank for a reason, e.g., the column is not relevant for certain complaints.

Question 2

The Deputy Mayor for Operations needs to understand if the data collection process has been consistent over time.

a) Investigate daily volume of 311 requests. Identify any unusual patterns, gaps, or anomalies in the data collection.
b) Now, investigate the times in which complaints were closed. Again, identify any unusual patterns, gaps, or anomalies.
c) Create a graphics that the city can use to present information on 311 request openings and closings over time. The city is particularly interested in day-of-week and week-of-year effects.

Question 3

The City Council has requested validation of complaint categories to ensure budget allocations are based on accurate data.

a) Analyze the 'Complaint Type' field for data consistency issues (e.g., similar categories with different names, unusual entries, etc.). Provide specific examples.
b) Propose a data cleaning strategy that would standardize these categories while preserving the granularity needed for city operations. Note that while it's great to handle all the data the city currently has, it would be even better if the strategy could be easily adapted when more data comes in.

Question 4

Response times are a key performance indicator for the Mayor's dashboard.

a) Calculate summary statistics for response times (if available) or time to closure. Include mean, median, standard deviation, and identify outliers.
b) The Mayor wants to promise "improved response times" in the State of the City address. Are there particular neighborhoods or areas in the city which are experiencing longer than typical response times?
c) Are there some agencies that are outperforming others in addressing complaints quickly? The Mayor wonders if longer response times is due to complaints being harder to handle. Conduct this analyze to inform the Mayor if some agencies are indeed offering exemplary services.

Part 2: Data Visualization & Insights

Question 5

The Mayor needs to present complaint patterns to the City Council next month.

a) Create a visualization showing the top 10 complaint types by volume. Your visualization should be "board-ready" - clear enough for the Mayor to use directly in the presentation. Include appropriate titles, labels, and formatting.

Question 6

The Office of Emergency Management wants to understand seasonal patterns to better allocate resources. There is reason to believe that some of these complaints may be due to weather conditions. For example, the roads may be flooded. Or, it may be so cold that pipes freeze. Unfortunately, the complaint data lacks these data features. However, some in the Mayor's office believes it's straightforward to collect historical weather conditions and merge into the analysis. Give this a try and create an appropriate graphic(s) that addresses how complaints (their volume and their content) relates to weather conditions.

Question 7

City Council members have requested district-level analysis for their constituencies.

Create a geographic visualization (this could be by borough, ZIP code, or community district) showing:

- The concentration of complaints across NYC
- Any clear disparities in service requests between areas

What three key insights about geographic inequality would you highlight to the Mayor? These must be included in the graphic.

Question 8

The Mayor's Office is particularly interested in developing a dashboard that allows staff to zoom in on an area and obtain details about complaints in that area. The office would like you to create a web-based graphic that achieves these goals.

Part 3: Combined Analysis & Strategic Recommendations

Question 9

The Mayor is preparing for a debate and expects questions about equity in city services.

Combine your geographic analysis with complaint types to identify:

- Which neighborhoods have the highest volume of unresolved complaints
- Whether certain types of complaints are concentrated in specific areas
- Whether complaints are originating from the city's actions, actions of people in the neighborhood, natural, or for other reasons
- Any patterns that suggest inequitable service delivery

Present your findings in a single, compelling visualization with accompanying insights.

Question 10

The Mayor's data team wants to build a predictive model for complaint volumes.

Based on your exploratory analysis:

- What features (variables) would be most valuable for prediction? Why?
- Test to see if these variables are correlated with response times.

Question 11

Executive Summary for the Mayor

You have 5 minutes with the Mayor tomorrow morning. Prepare a concise executive summary that includes:

- Three most critical findings from your analysis
- Two immediate action items

- One long-term strategic recommendation

Focus on insights that are actionable and feasible.

Submission Instructions

1. Submit all code, visualizations, and written responses
2. Ensure that all your code can be run with a single command and that it replicates all of your findings. Name your file.

Good luck!